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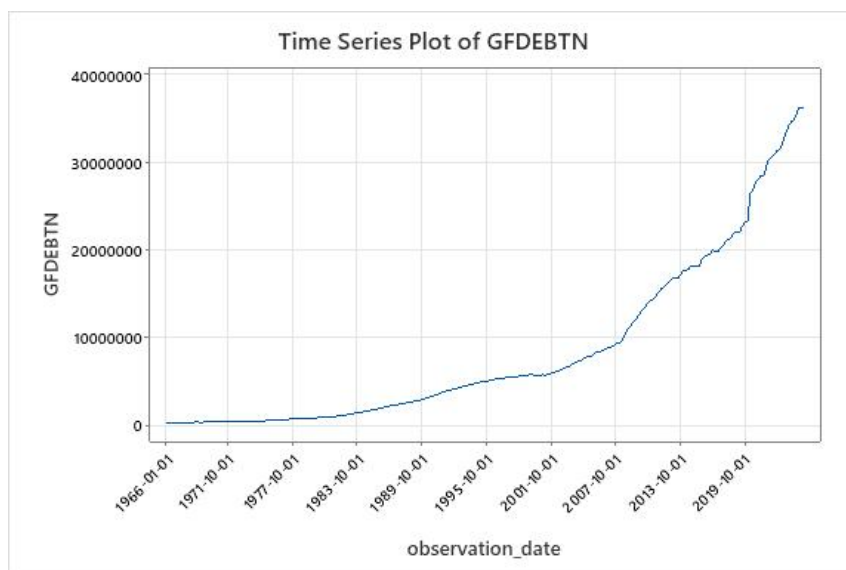
Forecasting Time Series Data, STAT-GB 6018

3 November, 2025

FORECASTING PROJECT 1

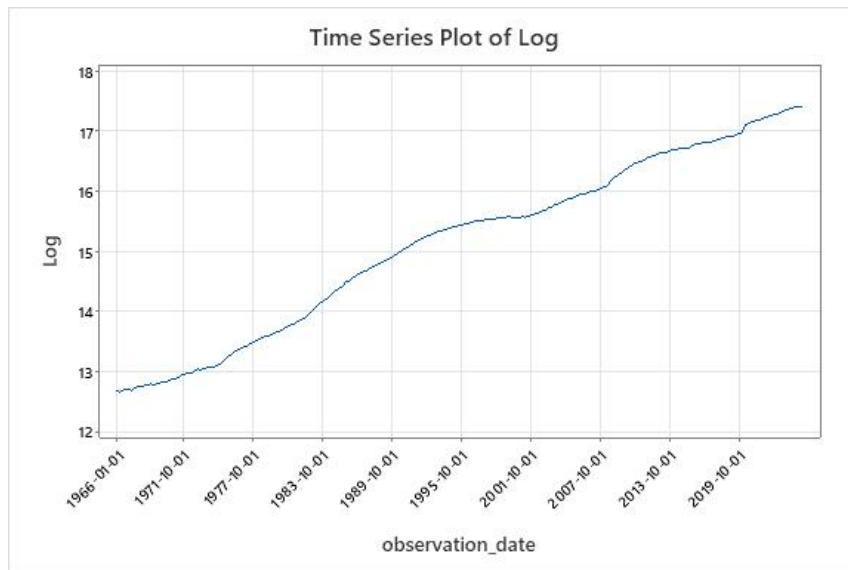
This project aims to model and forecast the quarterly Federal Debt: Total Public Debt in the United States using the ARIMA methodology. The dataset is obtained from the **Federal Reserve Bank of St. Louis (FRED)** under the series code **GFDEBTN** (Not Seasonally Adjusted), available at <https://fred.stlouisfed.org/series/GFDEBTN>. The data are measured quarterly in millions of dollars and cover the period from 1st January 1966 to 1st April 2025.

To begin, I plotted the original time series of GFDEBTN to visualize its general pattern.



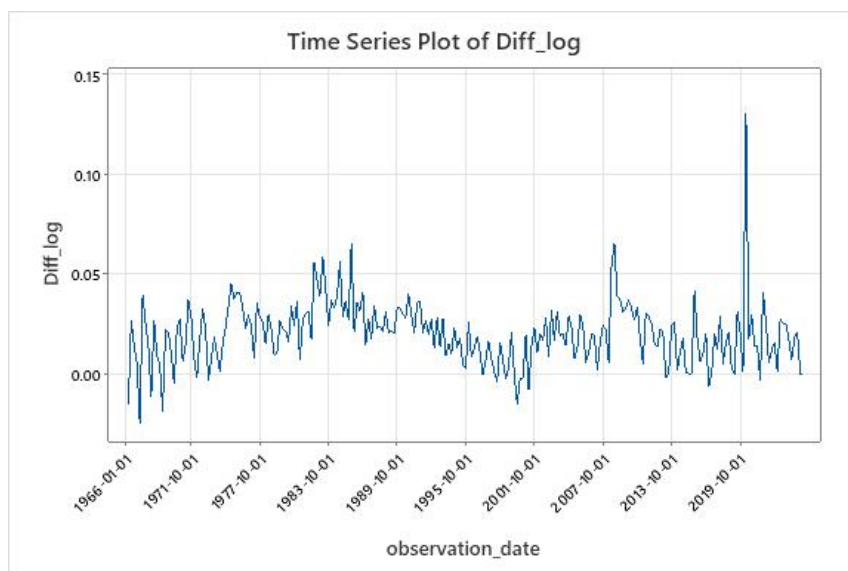
The series exhibits a clear upward trend with increasing volatility, resembling exponential growth. There are no visible seasonal oscillations, so seasonal adjustment is unnecessary. To stabilize the variance and smooth the model, we should convert the exponential trend growth into a linear trend. Thus, I decided to take the logarithm of the data to achieve this.

After taking log, the time series plot of Log_GFDEBTN is shown as follows:

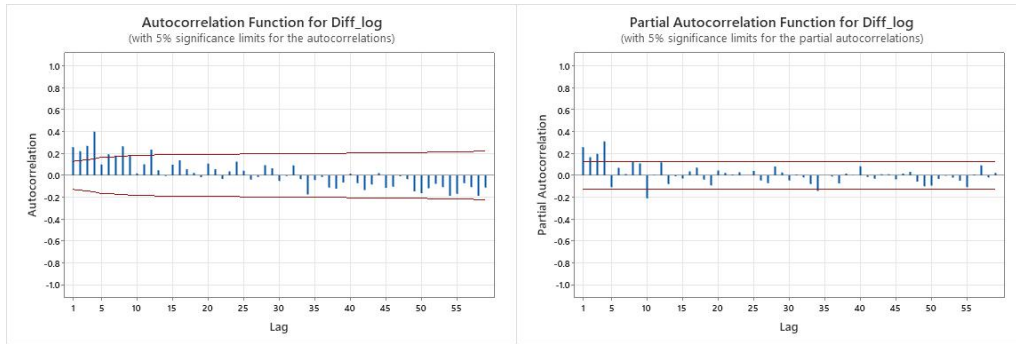


The time series plot of Log_ GFDEBTN represents an almost linear growth trend, which meets my expectation that the variability of the series becomes more uniform. Thus, **taking log is necessary** for my dataset.

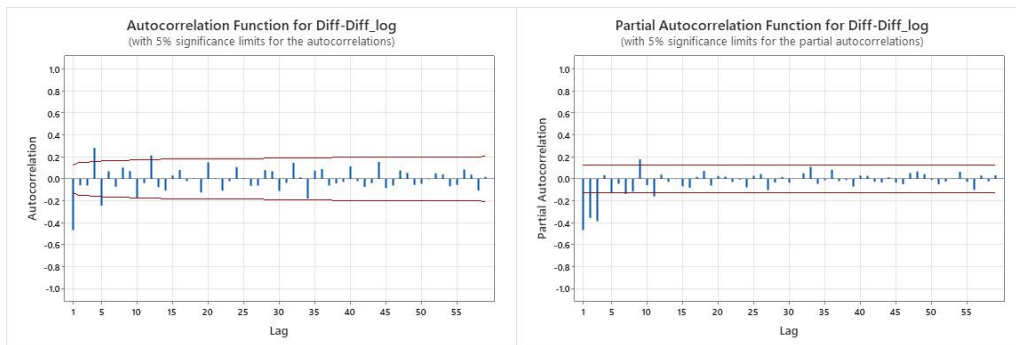
Nevertheless, the presence of a deterministic trend implies that the series is non-stationary. To achieve stationarity, I applied first differencing to remove the trend component.



This plot shows that I made the series fluctuate around a constant mean expect for one shock, suggesting that the first difference removed most of the systematic growth. After that, I plotted the ACF and PACF of the first differencing of Log_ GFDEBTN.



By observing the plot, the ACF of the first-differenced log series decays rapidly after lag 4, and most correlations lie within the 95% confidence bounds. The PACF also shows only the first 4 lags as significant. Therefore, the series seems to be stationary after the first difference. To be cautious, I took the second difference to examine its ACF and PACF.



The plots both shows a strong negative autocorrelations at lag 1, which means that it is overdifferenced, removing meaningful long-term strcture and increasing noise. To sum up, the series required **a first difference ($d = 1$)** to achieve stationarity.

To identify the most appropriate ARIMA model for the differenced Log series, I estimated several ARIMA models with $d=1$ and compared their $AICc$. Among the candidate models, the one with the lowest $AICc$ value was selected as the best ARIMA model for the Log dataset.

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 5, q = 5*	676.937	-1330.70	-1331.87	-1293.73
p = 4, q = 4	673.857	-1328.92	-1329.71	-1298.50
p = 5, q = 4	674.082	-1327.19	-1328.16	-1293.48
p = 4, q = 5	673.640	-1326.31	-1327.28	-1292.60
p = 3, q = 3	669.805	-1325.12	-1325.61	-1301.33
p = 4, q = 3	669.878	-1323.13	-1323.76	-1296.01
p = 3, q = 5	670.888	-1322.98	-1323.78	-1292.56
p = 3, q = 4	669.319	-1322.01	-1322.64	-1294.89
p = 5, q = 2	668.597	-1320.56	-1321.19	-1293.45
p = 5, q = 1	666.707	-1318.92	-1319.41	-1295.14
p = 5, q = 3	668.608	-1318.42	-1319.22	-1288.00
p = 4, q = 0	663.702	-1317.14	-1317.40	-1300.06
p = 4, q = 2	665.426	-1316.36	-1316.85	-1292.57
p = 4, q = 1	664.332	-1316.30	-1316.66	-1295.86
p = 5, q = 0	663.992	-1315.62	-1315.98	-1295.18
p = 2, q = 4	664.645	-1314.80	-1315.29	-1291.01
p = 1, q = 1	658.512	-1310.92	-1311.02	-1300.62
p = 2, q = 2	659.851	-1309.44	-1309.70	-1292.36
p = 3, q = 1	659.652	-1309.04	-1309.30	-1291.96
p = 2, q = 1	658.555	-1308.94	-1309.11	-1295.24
p = 2, q = 3	660.012	-1307.66	-1308.02	-1287.22
p = 3, q = 2	659.709	-1307.05	-1307.42	-1286.61
p = 1, q = 4	659.296	-1306.23	-1306.59	-1285.78
p = 3, q = 0	645.495	-1282.82	-1282.99	-1269.12
p = 2, q = 0	632.725	-1259.35	-1259.45	-1249.05
p = 0, q = 4	631.950	-1253.64	-1253.90	-1236.56
p = 0, q = 5	632.199	-1252.03	-1252.40	-1231.59
p = 1, q = 0	614.143	-1224.23	-1224.29	-1217.35
p = 0, q = 3	599.166	-1190.16	-1190.33	-1176.46
p = 0, q = 2	593.915	-1181.73	-1181.83	-1171.42
p = 1, q = 2	591.888	-1175.60	-1175.78	-1161.90
p = 0, q = 1	573.520	-1142.99	-1143.04	-1136.10
p = 0, q = 0	531.423	-1060.83	-1060.85	-1057.38

* Best model with minimum AICc.

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 4, q = 4*	677.490	-1334.01	-1334.98	-1300.30
p = 5, q = 4	677.461	-1331.75	-1332.92	-1294.77
p = 3, q = 3	673.817	-1331.00	-1331.63	-1303.89
p = 4, q = 5	676.760	-1330.35	-1331.52	-1293.37
p = 4, q = 3	673.595	-1328.40	-1329.19	-1297.98
p = 3, q = 4	673.373	-1327.95	-1328.75	-1297.53
p = 4, q = 1	671.178	-1327.87	-1328.36	-1304.08
p = 3, q = 5	674.004	-1327.03	-1328.01	-1293.33
p = 5, q = 0	670.675	-1326.86	-1327.35	-1303.07
p = 2, q = 5	672.718	-1326.64	-1327.44	-1296.22
p = 4, q = 0	669.406	-1326.45	-1326.81	-1306.00
p = 4, q = 2	671.184	-1325.74	-1326.37	-1298.62
p = 5, q = 1	671.179	-1325.73	-1326.36	-1298.61
p = 5, q = 2	671.555	-1324.32	-1325.11	-1293.90
p = 5, q = 3	671.697	-1322.42	-1323.39	-1288.71
p = 2, q = 4	669.506	-1322.38	-1323.01	-1295.27
p = 0, q = 5	668.230	-1321.97	-1322.46	-1298.18
p = 0, q = 4	666.742	-1321.12	-1321.48	-1300.67
p = 1, q = 5	668.491	-1320.35	-1320.98	-1293.24
p = 1, q = 1	661.556	-1314.94	-1315.11	-1301.24
p = 3, q = 1	663.342	-1314.32	-1314.68	-1293.88
p = 2, q = 2	662.969	-1313.57	-1313.94	-1293.13
p = 1, q = 2	661.828	-1313.40	-1313.66	-1296.31
p = 2, q = 1	661.744	-1313.23	-1313.49	-1296.15
p = 3, q = 2	663.797	-1313.10	-1313.59	-1289.32
p = 2, q = 3	663.032	-1311.57	-1312.06	-1287.79
p = 1, q = 4	659.296	-1306.23	-1306.59	-1285.78
p = 3, q = 0	657.036	-1303.81	-1304.07	-1286.73
p = 2, q = 0	652.301	-1296.43	-1296.60	-1282.73
p = 0, q = 3	651.316	-1292.37	-1292.63	-1275.29
p = 1, q = 0	648.961	-1291.82	-1291.92	-1281.52
p = 1, q = 3	651.886	-1291.41	-1291.77	-1270.96
p = 0, q = 2	648.806	-1289.44	-1289.61	-1275.74
p = 0, q = 1	646.865	-1287.63	-1287.73	-1277.33
p = 0, q = 0	640.571	-1277.09	-1277.14	-1270.21

* Best model with minimum AICc.

(With no constant)

(With constant)

The comparison results are summarized above. With no constant, the best model is ARIMA (5, 1, 5) with $AICc = -1330.7$. With constant, the best model is ARIMA (4, 1, 4) with $AICc = -1334.01$, which is smaller than -1330.7. Therefore, the best-fitting model based on AICc is **ARIMA (4, 1, 4) with constant term**.

The estimated ARIMA(4, 1, 4) model is given by the estimates of parameters as follows:

Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	-0.0146	0.0733	-0.20	0.842
AR 2	-0.1148	0.0733	-1.57	0.119
AR 3	-0.0414	0.0727	-0.57	0.570
AR 4	0.8422	0.0721	11.68	0.000
MA 1	-0.147	0.106	-1.38	0.169
MA 2	-0.335	0.102	-3.28	0.001
MA 3	-0.259	0.104	-2.48	0.014
MA 4	0.499	0.103	4.82	0.000
Constant	0.00643	0.00113	5.71	0.000

Differencing: 1 Regular

Number of observations after differencing: 237

The estimated model should be:

$$x_t = -0.0146x_{t-1} - 0.1148x_{t-2} - 0.0414x_{t-3} + 0.8422x_{t-4} + \varepsilon_t \\ - 0.147\varepsilon_{t-1} - 0.335\varepsilon_{t-2} - 0.259\varepsilon_{t-3} + 0.499\varepsilon_{t-4} + 0.00643$$

The estimation indicate that the AR(4) ($p = 0.000 < 0.001$) and MA(4) ($p = 0.000 < 0.001$) terms are highly statistically significant, implying that shocks to the series remain influential for approximately four quarters, producing a cyclical effect that repeats annually because the data is measured quarterly. The constant term ($p = 0.000 < 0.001$) is also statistically significant.

Then I examined the Ljung-Box statistics for lack of fit for the ARIMA (4, 1, 4) model with the constant term, which is:

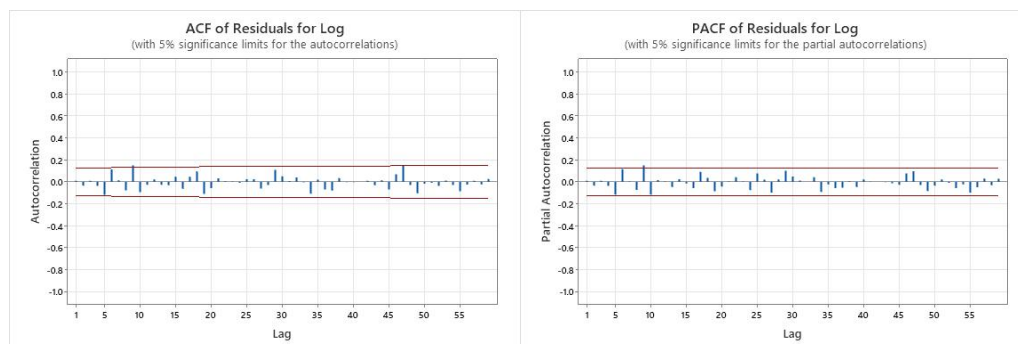
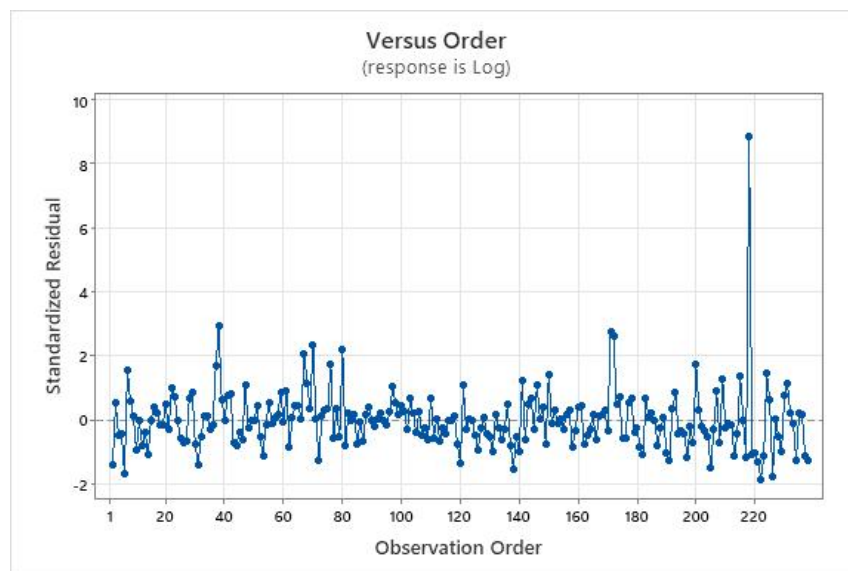
Modified Box-Pierce (Ljung-Box) Chi-Square Statistic

Lag	12	24	36	48
Chi-Square	16.80	25.95	36.56	48.73
DF	3	15	27	39
P-Value	0.001	0.039	0.104	0.137

According to the Ljung-Box test results in the table, the Chi-Square statistics at lag 12 and lag 24 are significant ($p < 0.05$) but become non-significant at higher lags. Given that the data are selected quarterly, these correspond to correlations over 3-year and 6-year horizons rather than annual effects. This indicates that the ARIMA model captures short-run dynamics well but may not fully account for longer-term cyclical

movements in the series. However, since the higher-order correlations are weak and dissipate at lga 36, the residuals might still be regarded as white noise. Thus, the model could still be seen as acceptable.

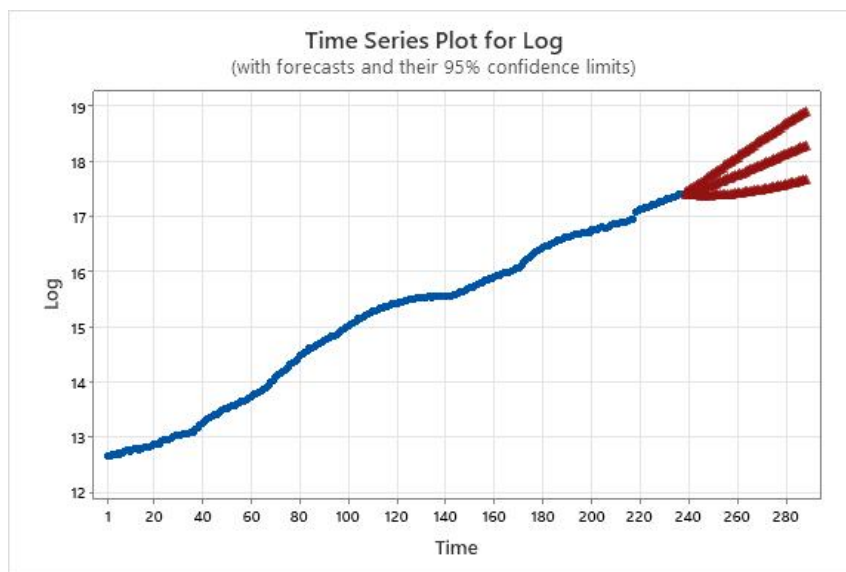
After estimating the ARIMA (4,1,4) model, I conducted a series of diagnostic tests. First, I plotted residuals versus Order, the ACF and PACF of residuals.



We can see that the residual fluctuate randomly around zero with no visible trend or structure, which ensure that systematic patterns have been removed. As for ACF and PACF plots, they show no significant spikes, and almost all correlations fall within the 95% confidence bounds. This pattern represents that the residuals are approximate white noise, meaning that the ARIMA (4, 1, 4) model has successfully removed the short-term in the original series and captured the short-run dynamics of the series.

Overall, the diagnostic results indicate that the residuals approximate a white-noise process. Therefore, the ARIMA (4, 1, 4) model can be considered statistically **adequate**.

Finally, by using the estimated ARIMA (4, 1, 4) model, I plotted the time series with forecasts for the next 50 quarters and 95% forecast intervals.



The forecasts show a smooth and linearly upward trend, consistent with the historical pattern. Also, the predicted values align closely with the recent observed data, representing that the model seems to be well-fitted, both in trend and in data.

The 95% prediction intervals widen gradually over time, which reflects that the uncertainty is accumulated over time. This pattern is a standard property of ARIMA model corecasts. At the same time, the intervals do not seem to be excessively wide, suggesting that the model is well-behaved and can still provide credible forecast of the long-term growth.

To conclude, the modeling and forecasting of the dataset GFDEBTN requires taking log due to its exponential trend. After appropriate transformation and differencing, I chose ARIMA(4,1,4) model with a constant term which is the best model due to its *AICc*. Residual diagnostics confirmed that the model captures short-run dependencies and leaves white-noise residuals. The forecast results are coherent with historical dynamics, with reasonable trend and appropriate ranges.

However, there exists some small problems during the process. A minor autocorrelation at multi-year lags suggests that long-term cycles may not be fully captured. But overall, the model's performance for medium-term forecasting is statistically solid and feasible.